The empirical recurrence for OEIS A231742, recovered from its tail

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Abstract

OEIS A231742 records "Empirical recurrence of order 87" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 105$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 89$. Its last coefficient is $c_{87} = -8 \neq 0$, so no further tail satisfies anything shorter; any order-87 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A231742 is "Number of nX3 0..3 arrays with no element less than a strict majority of its horizontal and vertical neighbors.". It has offset 1 and begins

$$16, 318, 4430, 60806, 784076, 9945132, 126926437, 1625269595, \ldots$$

The entry states:

> Empirical recurrence of order 87 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 06:39:17, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 105$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 105$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 87: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 87.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 210$ exact terms from $n = 2$ on, it returns order 87 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{87} c_i\, a(n-i),$$

c_{1}	40	c_{23}	433769036844	c_{45}	−68482998480250	c_{67}	2580369596484
c_{2}	−721	c_{24}	−1656803206918	c_{46}	220066801199110	c_{68}	2456185316788
c_{3}	8351	c_{25}	3437842999894	c_{47}	−40264056535166	c_{69}	267179175091
c_{4}	−71744	c_{26}	−3292192875226	c_{48}	−715378598171906	c_{70}	−443991677680
c_{5}	478442	c_{27}	−1072487395284	c_{49}	−326391835848142	c_{71}	−207099346909
c_{6}	−2531517	c_{28}	9069594646357	c_{50}	616141720164655	c_{72}	5887250032
c_{7}	10756674	c_{29}	−19857639419402	c_{51}	665315890636065	c_{73}	34026874910
c_{8}	−36829136	c_{30}	12265933248727	c_{52}	15614588185630	c_{74}	10525068419
c_{9}	101651121	c_{31}	24548518439778	c_{53}	−434428480448949	c_{75}	−732130779
c_{10}	−224247749	c_{32}	−44510794462917	c_{54}	−426177921201257	c_{76}	−1397475228
c_{11}	401911276	c_{33}	37720059523251	c_{55}	−66058412504806	c_{77}	−381125210
c_{12}	−679710121	c_{34}	17452394298885	c_{56}	298384515942894	c_{78}	−141273
c_{13}	1526442592	c_{35}	−126293736263711	c_{57}	270196388622583	c_{79}	29905664
c_{14}	−4705796798	c_{36}	55539104326036	c_{58}	−15487136945556	c_{80}	9484383
c_{15}	13226010464	c_{37}	117343908822586	c_{59}	−153277674800579	c_{81}	1106984
c_{16}	−27108695975	c_{38}	−86426762269695	c_{60}	−85579952535636	c_{82}	−172374
c_{17}	32744883694	c_{39}	69084866608755	c_{61}	11477805337773	c_{83}	−102122
c_{18}	8881419118	c_{40}	117041433267182	c_{62}	46642386092920	c_{84}	−22652
c_{19}	−136791085108	c_{41}	−321354828898553	c_{63}	27275659769786	c_{85}	−3000
c_{20}	337532358689	c_{42}	−228698389528094	c_{64}	−4643493392307	c_{86}	−232
c_{21}	−483437940773	c_{43}	343033252638837	c_{65}	−14336090083392	c_{87}	−8
c_{22}	269137804732	c_{44}	205872927928340	c_{66}	−5050091745696		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 89$, and it is the recurrence OEIS A231742 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{87} - c_1x^{87-1} - \cdots - c_{87}$. Its constant term is $-c_{87} = 8$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 87 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 87, so $\deg q = 87$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 87, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 87, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -8.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-87 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A231742.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.